DroneArjuna — MAVLink Message Headers Reference

Catalog of MAVLink message types used in the DroneArjuna Ground Control System (mavlink_manager.py, telemetry_processor.py, hf_link_adapter.py, mission_planner.py, com_bridge.py). Dialect: ArduPilotMega · pymavlink 2.4.41 · MAVLink2 framing.

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Link & Health Messages

HEARTBEAT *(ID 0)*

Sent by every MAVLink system at ~1 Hz to announce it is alive and to report its current autopilot type, vehicle type, and flight mode/arm state. It is the primary liveness signal used to detect link loss.

Key fields:

- type — vehicle type (quadrotor, fixed-wing, etc.)
- autopilot — autopilot class (e.g. ArduPilotMega)
- base_mode — armed/disarmed, custom-mode-enabled flags
- custom_mode — autopilot-specific flight mode id
- system_status — MAV_STATE (standby, active, critical, etc.)

Used in DroneArjuna: Used by mavlink_manager.py and health_monitor.py to detect connection loss and determine current flight mode/arm state for the fleet dashboard. Marked HIGH priority on bandwidth-constrained HF links.

STATUSTEXT *(ID 253)*

Free-text status/diagnostic messages emitted by the autopilot (e.g. 'Arming checks failed: GPS', 'EKF variance'), tagged with a severity level.

Key fields:

- severity — MAV_SEVERITY (emergency … debug)
- text — up to 50 chars of free text

Used in DroneArjuna: Surfaced directly to the operator as toast/log messages; LOW priority on constrained links since it is advisory, not flight-critical.

PARAM_VALUE *(ID 22)*

Response to a parameter read/write request, returning a single autopilot parameter's current value, type, and its index within the full parameter set.

Key fields:

- param_id — parameter name
- param_value, param_type
- param_count, param_index

Used in DroneArjuna: Used when reading/writing autopilot configuration parameters; LOW priority, generally only exchanged during configuration, not continuous telemetry.

RADIO_STATUS *(ID 109)*

Reported by the radio modem itself (e.g. SiK telemetry radio) rather than the autopilot, describing RSSI, noise floor, and remaining buffer space on the link.

Key fields:

- rssi, remrssi — local/remote signal strength
- noise, remnoise — local/remote noise floor
- txbuf — percent of transmit buffer free

Used in DroneArjuna: Explicitly whitelisted in mavlink_utils.py as an 'unsigned-allowed' message id (exempt from MAVLink2 message-signing requirements) since radio status often arrives unsigned from the modem hardware itself.

Telemetry Stream Messages

SYS_STATUS *(ID 1)*

Reports the general system state: battery voltage/current/remaining, onboard sensor health bitmasks, and communication drop-rate/error counters.

Key fields:
• onboard_control_sensors_health — bitmask of sensor health
• voltage_battery, current_battery, battery_remaining
• drop_rate_comm, errors_comm — link quality indicators
Used in DroneArjuna: Streamed at 2 Hz (via MAV_CMD_SET_MESSAGE_INTERVAL) for system/battery health monitoring.

GPS_RAW_INT *(ID 24)*

Raw GPS fix data direct from the GPS receiver, including fix type, satellite count, position, and horizontal/vertical dilution of precision.

Key fields:
• fix_type — 0=no fix … 3=3D fix, up to RTK
• lat, lon, alt — position (1e7 deg, mm)
• satellites_visible, eph, epv — fix quality
Used in DroneArjuna: Streamed at 2 Hz; used for GPS-lock status indicators and fix-quality warnings in the UI.

GLOBAL_POSITION_INT *(ID 33)*

Fused, filtered global position estimate (from EKF, blending GPS/IMU/baro), including altitude and velocity components.

Key fields:
• lat, lon, alt, relative_alt
• vx, vy, vz — ground velocity (cm/s)
• hdg — compass heading
Used in DroneArjuna: Streamed at 4 Hz; primary source for live map position/track and altitude readouts.

ATTITUDE *(ID 30)*

Vehicle orientation (roll, pitch, yaw) and angular rates, derived from the EKF.

Key fields:
• roll, pitch, yaw — radians
• rollspeed, pitchspeed, yawspeed — rad/s
Used in DroneArjuna: Streamed at 10 Hz (highest rate of all telemetry) to drive the attitude indicator / artificial horizon widget smoothly.

VFR_HUD *(ID 74)*

Classic 'heads-up display' metrics combining airspeed/groundspeed, climb rate, heading, and throttle percentage in one compact message.

Key fields:
• airspeed, groundspeed — m/s
• heading — degrees
• throttle — percent
• alt, climb — altitude and climb rate
Used in DroneArjuna: Streamed at 4 Hz for the HUD-style telemetry strip in the Fly workspace.

RC_CHANNELS *(ID 65)*

Raw RC receiver channel values (PWM) for all input channels, plus received signal strength.

Key fields:
• chan1_raw … chan18_raw — PWM microseconds
• rssi — received signal strength indicator
Used in DroneArjuna: Streamed at 2 Hz; used to display stick/RC input state and link RSSI. Deprioritized (suppressed) on constrained HF links.

BATTERY_STATUS *(ID 147)*

Detailed battery pack status beyond SYS_STATUS: per-cell voltages, consumed energy, temperature, and estimated remaining capacity.

Key fields:
• voltages[] — per-cell voltage array
• current_battery, current_consumed, energy_consumed
• battery_remaining — percent
• temperature
Used in DroneArjuna: Used for detailed battery panels/alerts; classified HIGH priority for link-budget purposes.

RAW_IMU *(ID 27)*

Raw accelerometer, gyroscope, and magnetometer readings direct from the IMU, prior to filtering.

Key fields:
• xacc, yacc, zacc — raw accelerometer
• xgyro, ygyro, zgyro — raw gyroscope
• xmag, ymag, zmag — raw magnetometer
Used in DroneArjuna: Available for diagnostics/vibration analysis; suppressed on bandwidth-limited HF links as LOW priority.

SERVO_OUTPUT_RAW *(ID 36)*

Raw PWM output values being sent to each servo/motor output channel — useful for verifying control surface / motor response.

Key fields:
• servo1_raw … servo16_raw — PWM microseconds output
Used in DroneArjuna: Available for actuator diagnostics; suppressed as LOW priority on constrained links.

NAV_CONTROLLER_OUTPUT *(ID 62)*

Internal navigation controller state: target bearing/distance, altitude error, and cross-track error relative to the current navigation target.

Key fields:
• nav_roll, nav_pitch, nav_bearing, target_bearing
• wp_dist — distance to active waypoint
• alt_error, aspd_error, xtrack_error
Used in DroneArjuna: Used to show distance-to-waypoint and navigation accuracy during missions; MEDIUM priority.

Command / Acknowledgement Messages

COMMAND_ACK *(ID 77)*

Acknowledgement/result of a previously sent COMMAND_LONG or COMMAND_INT, reporting whether the command was accepted, denied, temporarily rejected, or is still in progress.

Key fields:
• command — the MAV_CMD id being acknowledged
• result — MAV_RESULT (ACCEPTED, DENIED, UNSUPPORTED, FAILED, IN_PROGRESS)
Used in DroneArjuna: Marked CRITICAL priority — always delivered first/guaranteed on constrained links, since command confirmation (e.g. arm, takeoff, RTL) is safety-critical for the operator.

Mission Protocol Messages

MISSION_CURRENT *(ID 42)*

Reports the index of the waypoint/mission item the vehicle is currently navigating to.

Key fields:
• seq — index of the active mission item
Used in DroneArjuna: Streamed during active missions to highlight the current waypoint on the map; MEDIUM priority.

MISSION_ITEM_REACHED *(ID 46)*

Sent once when the vehicle reaches a given mission item, signalling progression to the next waypoint.

Key fields:
• seq — index of the mission item just reached
Used in DroneArjuna: Used to advance the mission-progress UI and trigger waypoint-reached notifications; deprioritized (LOW) on constrained links since it's event-driven, not continuous.

MISSION_REQUEST *(ID 40)*

Legacy mission-protocol message: the vehicle requests a specific mission item (by sequence number) from the GCS during mission upload.

Key fields:
• seq — requested mission item index
• mission_type — MAV_MISSION_TYPE (mission/fence/rally)
Used in DroneArjuna: Part of the mission upload handshake in mission_planner.py when pushing a flight plan to the autopilot.

MISSION_REQUEST_INT *(ID 51)*

Newer integer-coordinate variant of MISSION_REQUEST, preferred for precision when transferring waypoints with fixed-point lat/lon.

Key fields:
• seq — requested mission item index
• mission_type — MAV_MISSION_TYPE
Used in DroneArjuna: Used interchangeably with MISSION_REQUEST depending on what the connected autopilot firmware supports.

MISSION_ACK *(ID 47)*

Final acknowledgement that closes out a mission upload/download transaction, reporting success or a specific mission-protocol error.

Key fields:
• type — MAV_MISSION_RESULT (ACCEPTED, ERROR, INVALID_SEQUENCE, etc.)
• mission_type — MAV_MISSION_TYPE
Used in DroneArjuna: Checked explicitly in mission_planner.py (msg.get_type() == 'MISSION_ACK') to confirm a mission upload completed successfully before reporting status back to the operator.